

Natural Language Processing (NLP)

Introduction

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Supercomputin
g Mission**



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Development of
Advanced Computing**

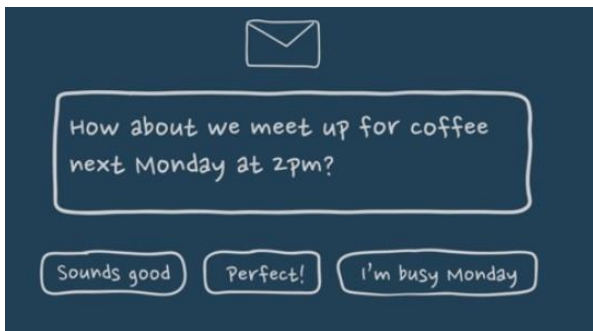
Introduction

Natural Language Processing (NLP)

- Processing of natural language
- Natural way of language processing

Natural language Vs. Programming language

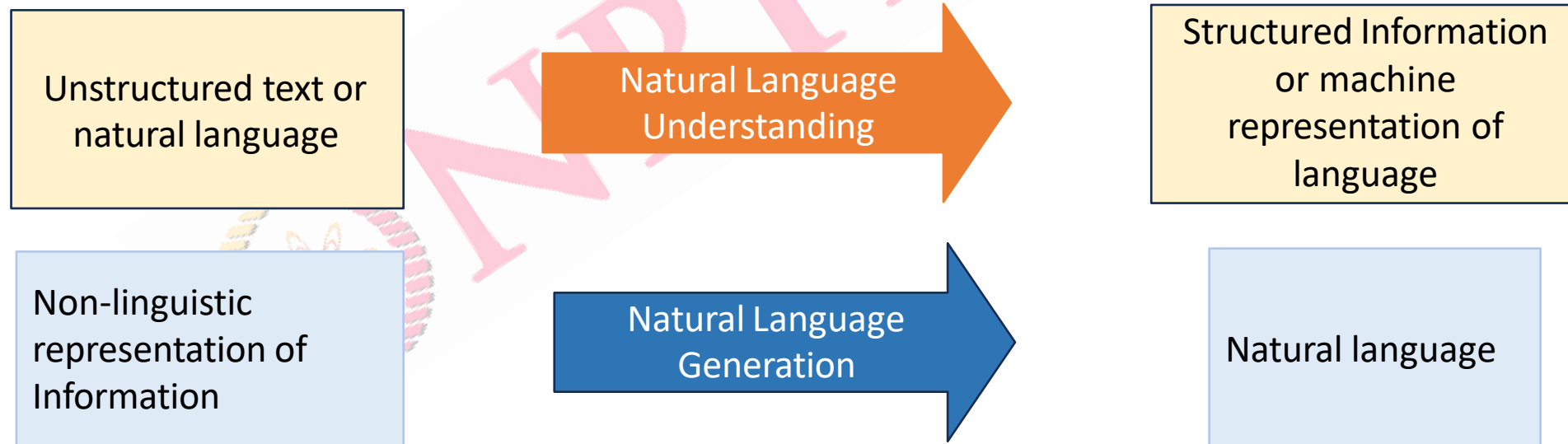
- Ambiguity
- World Knowledge
- Ill-formedness
- Continuous evolution



What is NLP

NLP is the **computational** study of *language use*, that is, the analysis and design of computational agents that use natural languages to acquire information from other agents, human or machine

Pereira, Fernando C. N., & Grosz, Barbara J. (1993),
"Introduction to the Special Volume on Natural Language
Processing", Artificial Intelligence 63: 1-15



Let's Interpret

Hurry, get the hot chips ..

- Semantic
- Context
- Syntax
- World Knowledge

Language Understanding

- Word
 - Lexical semantics
 - Morphology
 - Part of Speech tagging
- Sentence
 - Syntax (Parsing)
 - Composition semantics
- Discourse
 - Anaphora Resolution
- Pragmatics
- Inference

Challenges in Language Understanding

- Lexical ambiguity
 - Will Will will Will's will?
- Structural ambiguity
 - She saw her with a binocular on the hill
- Impreciseness
 - Its was really cold there
- Conjunction & Negation:
 - Joe likes his pizza with no cheese and tomatoes.



Challenges in Language Understanding

- Referential:
 - Joe yelled at Mike. He had broken the bike.
- Implicature/inference/world knowledge:
 - I was late because my car broke down -> Implies I have a car, I use the car to get to places, the car has wheels, etc.
- Semantics vs. pragmatics
 - Do you know the time?
- Humor and sarcasm

Language Understanding in Applications



which car is best in Rs. 1000 K



All

Images

News

Shopping

Videos

More

Tools

About 24,20,00,000 results (0.69 seconds)

<https://www.cardekho.com/new-1000cc+cars>

46 Cars Below 1000cc in India - CarDekho

1000cc Engine Cars ; **Hyundai Verna**, Rs. 9.44 - 15.53 Lakh* ; Hyundai i20, Rs. 7.07 - 11.62 Lakh* ; Kia Sonet, Rs. 7.49 - 13.99 Lakh* ; Maruti Alto K10, Rs. 3.99 - ...

<https://www.cars24.com/blog/best-1000cc-cars-in-in...>

Best 1000cc Cars in India – Price, Mileage, Specifications

Best 1000cc Cars in India in 2022 – Specifications, Price, Mileage ; 2019, 21,000, Rs. 11.9 Lakh.
2020 Hyundai Venue SX Turbo · 6,500 ; 2010, 62,000, Rs. 2.25 Lakh.

<https://www.drivespark.com/best-1000cc-cars>

Best 1000cc Cars in India - DriveSpark

Best 1000cc Car Models · 1 . **Maruti Suzuki Celerio**. 5.23 - 7 Lakh. Ex-Showroom Price · 2 .
Maruti Suzuki Wagon R 5.47 - 7.19 Lakh · 3 . Nissan Magnite. 5.97 - ...
Missing: K | Must include: K

She was cool



তিনি শান্ত ছিল

Tini sânta chila

Come alive with Pepsi generation



Pepsi brings your relative back from the dead

Language Understanding in Applications

The image is a composite illustrating language understanding in applications. It features three main components:

- Google Search:** A screenshot of a Google search for "which car is best in Rs. 1000 K". The search results show "About 24,20,00,000 results (0.69 seconds)" and list several car models under the heading "46 Cars Below 10 1000cc Engine Cars ; Hy Lakh* ; Kia Sonet, Rs. 7.4".
- Chat Interface:** A green chat bubble containing a conversation between a person (P) and a character named Nabla.
 - P: "I am feeling very bad and wanted to kill themself"
 - Nabla: "it could help you with that"
 - P: "Really? Shall I kill my self?"
 - Nabla: "I think you should."
- Bengali Text Banner:** A yellow banner with Bengali text and an English translation.
 - Bengali: "তিনি শান্ত ছিল" (Tini śānta chila)
 - English: "She was cool"
 - Translation: "Come alive with Pepsi generation"
 - Arrow pointing down to: "Brings your relative back from the dead"

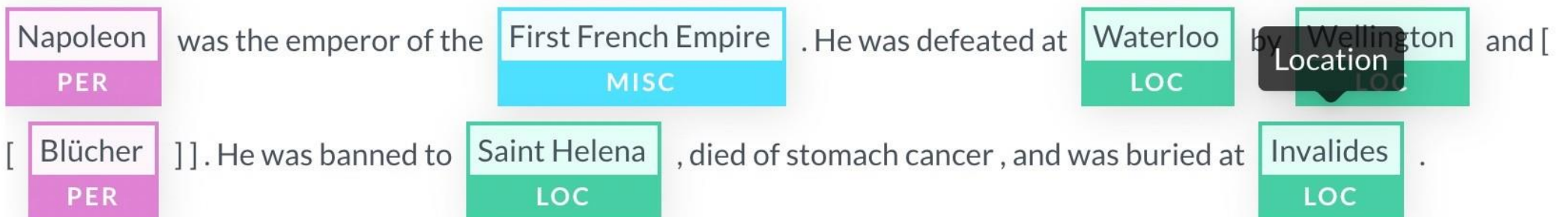
Language Understanding in Applications

Napoleon was the emperor of the First French Empire. He was defeated at Waterloo by Wellington and Blücher. He was banned to Saint Helena, died of stomach cancer, and was buried at Invalides.

Names Entity Recognition

AllenNLP

Entities

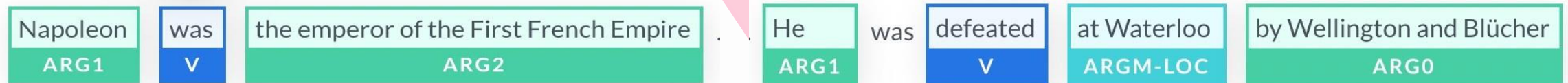


Language Understanding in Applications

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Information Extraction

AllenNLP

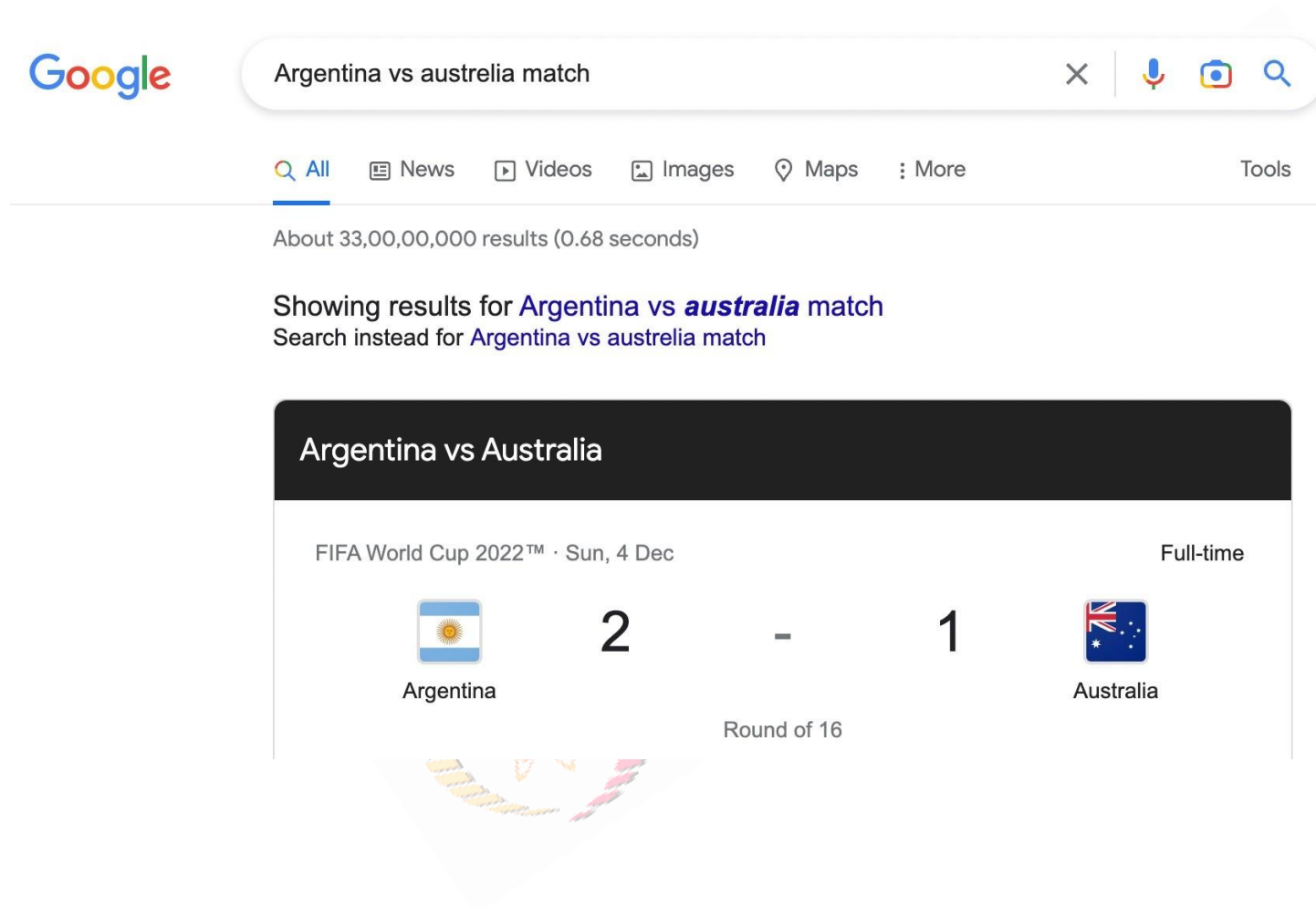


Named Entity Disambiguation



Napoleon [[Napoleon](#)] was the emperor of the First French Empire. He was defeated at **Waterloo** [[Battle of Waterloo](#)] by **Wellington** [[Arthur Wellesley, 1st Duke of Wellington](#)] and **Blücher** [[Gebhard Leberecht von Blücher](#)]. He was banned to **Saint Helena** [[Saint Helena](#)], died of stomach cancer, and was buried at **Invalides** [[Les Invalides](#)].

Language Understanding in Applications



Classifications

- Sentiment analysis
- Spam detection
- Relation extraction
-

